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Chapter 1

Introduction

Tonal languages are estimated to constitute 60 to 70 percent of the world’s languages (Yip, 2002), making tonal phonology a critical and unique subfield within phonology. One key characteristic distinguishing tonal phonology from segmental phonology is its autonomy: tones can operate independently of the segments with which they are associated. This autonomy has been captured by Autosegmental Phonology which uses a multi-tiered structure for tonal representation (Goldsmith, 1976) and gives more accurate predictions and explanations than linear phonology (Clements and Goldsmith, 1984; Odden, 2013). Computationally, processes in segmental phonology are characterized as sub-sequential and can be handled by a deterministic transducer (Heinz and Lai, 2013) while some processes in tonal phonology such as unbounded tone spreading are found to be more complex and non-deterministic (Jardine, 2016, 2017). However, beyond the formal properties of tones (Bird and Klein, 1990; Coleman and Local, 1991; Kornai, 2018; Jardine and Heinz, 2015), limited work has been done on tonal pattern learning, and more importantly, little effort has been made to capture the multi-linear properties of tones in the computational linguistics field (Albro, 1994; Kornai, 2018), despite their prevalence in the theoretical world.

In the proposed dissertation, I aim to provide a computational expression using formal logic to make tonal patterns and processes applicable for computational modeling and learning. While the previous attempts in tonal learning use a string representation (Chandlee, 2019; Oakden and Chandlee, 2020), this work plans to develop a multi-linear structure faithful to linguistic representation that captures the autonomy of tones. Secondly, autosegmental phonology has achieved a great number of success in African tonal languages (Clements and Goldsmith, 2010; Hyman, 2009), but far less research has explored Asian tonal languages which are known for having intricate tone-segment interactions. Incorporating these inter-

acting features into the modeling is another goal of this project. Finally, it is noteworthy that one possible reason that tonal phonology has not obtained the same level of discussion in computational linguistics is due to limited tools and resources. No algorithm to date has been able to effectively learn the input-output mapping of autosegmental representations, which are intrinsically graphs. Besides, tonal languages are often transcribed without tonal specifications (as in African language dictionaries) or are transcribed in their underlying representations (as in Mandarin Chinese). For those languages that do have sufficient data, there are usually varying representations or models (Bao, 1999; Yip, 1989; Wang, 1967). The proposed work aims to clarify these issues and hopes to develop more tools for tonal linguists to use in learning tonal patterns and processes. These goals will help to gain more insights into tonal phonology and its learnability, provide implications for theoretical generalizations, and get a better understanding of underrepresented languages with limited data.

1.1 Thesis Structure

Chapter 3, “Computation of Tonotactic Patterns,” is ongoing and is planned to be completed by the end of this year. Building on my second qualifying paper, this work involves expanding the existing algorithm to encode syllable weight and word boundaries in Hausa, and then apply it to Japanese Nouns. Corpora for both languages have already been gathered and are ready to be implemented into the modified BUFIA pipeline. My goal is to complete both experiments by the end of this semester and write up the chapter.

Chapter 4, “Interaction between Tone and Other Features” will be carried out roughly from the winter spanning over the spring semester of 2025. The winter time will be used for research, reading, and writing for the backgrounds. From February to April, I will start experimenting with perhaps two case studies from Shanghai Chinese, Mandarin, or Fuzhou. The plan for Chapter 3 depends on the progress of the two experiments.

Chapter 5, “Tonal Process Learning,” will take the longest to complete, as there is limited research on the computation and modeling of this topic. I will need additional time to learn new material and conduct further reading. In the timeline, I have allocated May to September for this chapter, but I have also reserved October and November for both writing and addressing any unresolved issues that may arise during the completion of this chapter.

The introduction chapter provides an overview, including three main points. First, tonotactic well-formedness and tonal processes differ from those at the segmental level, making it reasonable to adopt multi-tiered structures rather than string representations. Second, from

a learning perspective, computational tonal phonology is still underexplored. Most existing studies use string representations; many focus on African languages, likely due to the convenience of obtaining tonal data; and very few learning models have been developed for either tonal processes or tonotactic patterns. These research gaps raise important questions regarding representation, patterns, and learning. These questions will be the main focus of this dissertation and will be addressed in the introduction chapter. Chapter One will also include logistical details, such as an outline of the dissertation.

Chapter 2 will serve as a Literature Review chapter and it will mainly cover two broader aspects around tonal phonology. First, as in the overview in this proposal, I will review the issues about how representation, grammar, and typology of tonal languages. Second, I will discuss how previous work regulates the formal properties of tonal representations using model theory or other mathematical methods.

As the first major part of the dissertation, this chapter will investigate the tonotactic grammar of tonal languages by examining forbidden structures in all surface-true forms. In this chapter, I choose Hausa and Japanese as case studies to represent two typologically different tonal languages. Hausa uses a level tone system where H and L contrast and Japanese employs a pitch accent system. These patterns are argued to be local and their grammar can be learned by searching for the negative literal conjunctions. My goal is to represent the tonal patterns in model theory, apply a bottom-up learning algorithm to learn the grammar, and lastly compare the constraints found by the algorithm with previously reported generalizations, and see to what degree they match with each other.

The frequently discussed tonal patterns in Hausa are generalized as No Rising Contour, Avoid LL sequence, No initial Contours, and No three-tone contours, but it is unclear to what extent these rules can explain the data, and there are many counterexamples against these generalizations. In my QP2, I used a small Hausa corpus, from which I translated the orthographic representations into ARs and implemented BUFIA on it. The experiment revealed 26 distinct ARs among 664 surface-true Hausa words and identified 7 AR structures (syllable number ≤ 3 , tone number ≤ 3) missing from the data, which constitute the constraints in the tonal grammar. The computationally discovered constraints either match the ones previously proposed or provide more specific generalizations that better account for the data.

1.2 Motivations

Effective tone modeling requires choosing appropriate representations and grammar. The motivation behind this work stems from the most common questions in tonal phonology, which address tonal representation, grammar, and learning.

1.2.1 Representation

The first question is what phonological representation should be used for tones. Under the influence of the classic framework of *Sound Patterns of English* (Chomsky and Halle, 1968), where phonological strings were divided into discrete feature matrices, tones were likewise treated as intrinsic properties of segments and represented with features indicating pitch ($\pm H$) and contour shape ($\pm$ contour). Tonal changes are the result of changing feature values and insertion/deletion of segments. Contour comes into form due to two opposite pitch features are adjacent to each other. This approach, however, was soon argued to be inadequate for predicting correct attested patterns, and it failed to capture the commonly observed patterns among tonal languages (Hyman, 2008; Odden, 2013). Therefore, “remarkably little insight was produced by the SPE framework into the behavior of tones” (Clements and Goldsmith, 1984, p.8).

Abundant evidence from African tonal languages shows that when considering tonal forms, tonal units and their associations with timing units also need to be taken into account. For example, languages with “restricted melodies,” such as Mende (Leben, 2017), have a fixed set of five melodies to any form regardless of the number of syllables of the form. These melodies “must be abstracted away from the TBUs on which they are realized” (Voorhoeve, 1973; Hyman and Schuh, 1974). To capture this, Autosegmental Theory (Goldsmith, 1976; Clements, 1980) enriches the geometry of phonological representation by proposing a multilinear structure: tones are represented on a separate autonomous tier and are related to segments on the adjacent tier through association lines governed by specific well-formedness conditions (WFC) (see also Clements and Goldsmith (2010)). Common tonal patterns such as contours or tone plateaus can be explained as multiple tones connecting to the same segment or one tone spanning across multiple segments. The insertion or deletion of one tier does not necessarily trigger the same operation on the other tier. The temporal rearrangements between tones and TBUs are handled by changing the association lines between two tiers. Autosegmental Theory has proven especially useful in explaining tonotactic patterns and tonal processes where linear phonological models often struggle (Hyman, 2011; Odden, 2013).

Feature matrices and Autosegmental Representations (ARs) are not the only approaches to tonal analysis. [Inkelas and Shih \(2013\)](#) introduced *Q-theory*, initially designed to account for complex diphthongs, which later expanded to contour tones. Under this framework, each tone (or segment) can be divided into three temporally ordered, quantized subsegments. This is motivated by the observation that most complex tones do not exceed three tonal levels and very rarely have four-tone contours ([Shih and Inkelas, 2013](#)). However, this approach has been argued to be logically equivalent to ARs in model theory and logical interpretations ([Jardine et al., 2021](#)). [Kornai \(2018\)](#) also compared different ways of encoding string-based representations of ARs. Similarly, the two tonal models proposed by [Yip \(1989\)](#) and [Bao \(1999\)](#) are also shown to be notationally equivalent using formal logic ([Oakden, 2020](#)).

Given that varying models and representations are argued to be logically equivalent, or at least translatable, formal logic is possible to represent their internal structure and facilitate learning explicitly. This is the motivation for this study.

1.2.2 Grammar

Besides representations, the second question in tonal phonology concerns the type of grammar that should be used. The first distinction is whether the grammar is rule-based or constraint-based. Take the classic tonal mapping problem for instance. In a rule-based approach, tonal mapping requires a globally evaluated direction (leftward vs. rightward), along with other parameters such as morphological category and tone quality ([Goldsmith, 1976](#)), but this cannot account for those two-edge effects, and sometimes leads to wrong predictions. [Zoll \(2003\)](#) argues that the directionality needs to be reconsidered, and proposes the optimal tonal mapping where directionality can be handled by the interaction between universal violable constraints: a morphological direction constraint ALIGN and two quality-sensitive markedness constraints CLASH and LAPSE.

[Jardine \(2017\)](#) argues that the grammar of tonotactic well-formedness consists of language-specific, inviolable local constraints. These constraints can be expressed locally over ARs, which are intrinsically graphs. A well-formed AR structure can be viewed as a graph that does not contain any forbidden subgraphs. Using Model Theory, each AR graph can be abstracted into a mathematical model with a unique signature to indicate its distinct internal structure. This method captures the locality and nonlinearity makes it possible to learn tonotactic patterns over ARs, and also explains the North Karanga Shona “edge-in” pattern that Optimal Tonal Theory fails to generalize. More importantly, this approach of searching subfactors to get grammar can be learned through The Bottom Up Factor Inference

Algorithm (BUFIA, [Chandlee et al., 2019](#)).

1.3 Contributions

fukc you fuck you

Chapter 2

Tones: Theory and Computation

Languages with tonal distinctions have been studied as a more or less separate subfield known as tonal phonology. This division reflects a natural difference between the nature of tone and that of segments, as evidenced by a large body of empirical evidence from Asian and African languages. Rather than being inherent properties of vowels, tones may change independently of the segments that bear them. This property challenges traditional linear rule-writing approaches, which provide a cumbersome means of capturing tonal generalizations. Consequently, any discussion of tone necessarily involves a discussion of its representations, ranging from accent marks and feature-based representations to phonetically based notational systems(?). These approaches can be found throughout both the African and Asian tonal phonology literature. Among them, Goldsmith’s (1976) autosegmental theory has become the most influential and widely adopted framework. The first half of this chapter provides an overview of the major approaches to tonal representation and typology.

The issue of representation further poses important requirements for computational modeling. Since the introduction of autosegmental theory by Goldsmith (1976), there has been a clear shift in theoretical phonology toward structurally enriched representations. In contrast, computational approaches have been slower to adopt such representations. Works by Jardine (2016); Chandlee and Jardine (2019); (?); Oakden and Chandlee (2020) argue that linguistically meaningful representations can reduce computational complexity. These representations include autosegmental graphs (Jardine, 2016), metrical structures [CITE: lamount], and feature-based representations for segmental learning [CITE: magda]. These proposals, together with their computational implementations, highlight an important direction for computational modeling: the use of linguistically informed representations to improve interpretability, data efficiency, and linguistic adequacy through the development of

small, specialized language models. The second half of this chapter reviews computational approaches to tonal phonology, including finite-state transducers and other computational models. It also discusses the challenges and opportunities involved in developing computational models that effectively capture the complexities of tonal phonology.

2.1 Representations of Tone

Estimates of the proportion of the world’s tonal languages vary from 42.9% (?) Yip (2002), for example, suggests that as many as 60–70% of the world’s languages may be tonal, whereas Maddieson (2013) estimates the proportion to be closer to 40%. This discrepancy reflects differences in language sampling as well as in the criteria used to distinguish tone languages from languages with pitch accent or marginal tonal contrasts. Nevertheless, tone languages have a non-even geographical distribution: they are especially widespread in Africa, Central America, and East and Southeast Asia, while occurring more sparsely in Eurasia, North America, and Australia (Welmers, 1962; Maddieson, 2013; Kao, 2017; Gordon, 2016).

Since tonal languages are widespread across different geographic regions, different conventions for representing tones have been developed to accommodate the languages in questions. Yip (2002) classified three main notational traditions for tones: Africanist, Asianist, and Central Americanist. I group the latter two together for the present discussion since both use numerical notation, although they differ in how the numbers correspond to pitch levels and contours.

2.1.1 Letter System

An Africanist convention is to represent tone using accent marks. Tones are commonly marked with diacritics, even when they are not represented in a language’s everyday orthography. Table ?? shows a basic system.

Table 2.1: Common Africanist tone notation

Tone	Symbol	Letter
High	á	H
Low	à	L
Mid	ā	M
Falling	â	F
Rising	ǎ	R

It should be noted that Africanist notation varies across linguists and languages. First,

the unmarked tone in a two-tone system may be left without a diacritic; similarly, the mid tone is often unmarked in a three-tone system. Conversely, the macron ($\bar{a}$) is also used to indicate vowel length in some literature. **[CITE: Provide examples of unmarked-tone conventions in several African languages]**

Second, the notation of contour tones may be ambiguous, particularly in languages in which contour tones have greater distributional flexibility. As shown in Table ??, a falling tone may be written with a circumflex on the nucleus vowel ($\hat{a}$), on a long vowel ($\hat{a}:$), or on one element of a complex nucleus ($\hat{a}i$, $\hat{a}n$). Additionally, contour tones can emerge in the surface form as a morphological or phonological processes, even though the language does not have underlying contour form. One practice is to distinguish underlying contour from derivational contour with F vs HL and R vs LH. but this praxirix is not consistnetly used. therefore, contour tones may alternatively be represented as ($\acute{a}\grave{a} = \hat{a}:$). In languages that permit both concave and convex contour tones, circumflex and caron diacritics alone may be insufficient to represent more complex contours unambiguously.**[CITE: Mende tonal notation system]**

Table 2.2: Common Africanist tone notation

Tone	Symbol	Letter
High	$\acute{a}$	H
Low	$\grave{a}$	L
Mid	$\bar{a}$	M
Falling	$\hat{a}, \hat{a}:, \acute{a}\grave{a}$	F/HL
Rising	$\check{a}$	R/LH
Fall-rise	˘	FR/HLH
Rise-fall	$\grave{\text{a}}$	RF/LHL

Although these rules describe the observed mappings, they obscure the generalization that the tonal components are preserved while the number of tone-bearing units is reduced. A representation that decomposes contour tones into sequences of level tones captures this generalization more directly.

2.1.2 Phonetic-Based Notation

In the Asianist tradition, tone is commonly represented using Chao’s five-level system (Chao, 1930). This system has the following properties:

- 1. The natural pitch range of a speaker’s normal voice is divided into five levels, with 1 representing the lowest pitch and 5 the highest.

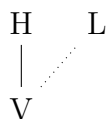
2. A sequence of digits indicates the starting point, any turning point, and the endpoint of a pitch contour. For example, 55 represents a high-level tone, 35 a rising tone, and 214 a falling-rising tone.
3. A toneless or neutral-tone syllable may be represented by 0.
4. An entering-tone syllable—that is, a syllable ending in a stop coda—is conventionally represented by a single digit.

2.1.3 Feature Theory

Feature theory is another option for describing tones. H, L, R, F can be presented as [+H,-contour], [-H,-contour], [-H,+contour], [+H,+contour]. In other words, R and F will be a natural class of [+contour] and L and H will be [-H]. However, this system is also inadequate for several reasons.

Firstly, this theory fails to produce the correct rules for patterns that do exist, and also produces unattested patterns. Example ?? lists two common rules when a H tone becomes a R after {L,F}, or a L tone turns into a F after {H,R}, and there is no possible way to describe the process by using [contour] feature because {L,F} or {H,R} does not form a natural class. Besides, the theory can also predict unattested patterns in Example ??.

As can be seen, this could be simply handled by ARs by adding an association line between L and H to the segment:



Another phenomenon which argues for the autosegmental representation of tone is across-the-board effects, which refers to the effect that a tone change applies to the whole domain rather than just its local neighbor. As shown in Example ?? from Shona, a H tone prefix will trigger a series of L tone to change into H. However, if more prefixes stack together, the melody will be in tone alternation, which would be extremely difficult for feature theory to capture accurately.

2.1.4 Q-Theory

2.1.5 Autosegmental Theory

Autosegmental theory, proposed by Goldsmith (1976), proposes that tone be represented autosegmentally in separate tiers from segments, but linked to them by association lines, and the association between tone and tone-bearing unit (TBU) is governed by a set of wellformedness conditions.

The foremost reason in support of autosegmental representation over linear theory is that latter struggles to capture the autonomy property of the tonal processes. One of the most common process in tonal phonology is tone preservation which means the deletion of a vowel segment does not lead to the deletion of the tone carried by the vowel. Example ?? from Yekhee (?) demonstrates the merger of two tones onto one syllable when the one of the tone-bearing unit is deleted after compounding. Linear rules in a string representations in (??) have two problems. First, the rule will over generalizing for all H and L tone merger when it is not; Besides, there is no formulation of the syllable deletion which contour formation was induced. In addition, new tones need to be introduced into the tone inventory. The contour formation process can be more simply handled in a representation of ARs as in (??)

All these above processes can be handled more elegantly by modifying association lines with tones and vowels by using ARs. (draw according examples below).

Lastly, autosegmental theory is advantageous is that it captures the nature of tonal process more accurately. Spreading is one of the most common process, which a tone unit spread over a domain of one or several segments. This process will be generalized as copying if we use a linear theory. For example, in Cameroon ?, the H tone spreads onto the L verb root and thus producing a HL as in (??). A rule format of (??) generalizes the H tone spreading process as a H tone copying or tone addition in the second syllable, although it is simply a H tone expands its domain.

Chapter 3

Formal Background

3.1 Formal Language Theory

3.2 Tonal Representations

3.3 Tonal Transduction

3.4 Learning Algorithm

Chapter 4

Tonotactics

Chapter 5

Tonal Processes

Chapter 6

Tone-segment Interaction

Chapter 7

Discussion

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